Name: ______________________

QUADRATICS, (COMPLETING THE SQUARE)

Date: _________________

LO: I can write a quadratic expression in completed square form.

What is completing the square?

Completing the square rewrites $ax^2 + bx + c$ in the form $(x + p)^2 + q$. This reveals the vertex (turning point) of the parabola.

Strategy: Halve the coefficient of x to find p, then subtract p^2 and adjust the constant to find q.

Quick examples

✓	$x^2 + 6x + 5 = (x + 3)^2 - 4$	half of 6 is 3, 5 - 9 = -4
✓	$x^2 - 4x + 1 = (x - 2)^2 - 3$	half of -4 is -2, 1 - 4 = -3
✓	$x^2 + 8x = (x + 4)^2 - 16$	no constant: q = -16
✗	$x^2 + 6x + 5 = (x + 6)^2 + 5$	must halve the coefficient
✗	$x^2 + 6x + 5 = (x + 3)^2 + 5$	forgot to subtract 9

Key idea

$x^2 + bx + c = (x + b/2)^2 + c - (b/2)^2$

$$x^2 + bx + c = (x + b/2)^2 + c - b^2/4$$

Halve b for the bracket; subtract the square for q

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – positive coefficient of x

Write $x^2 + 8x + 10$ in completed square form.

Half of 8 is 4, so bracket is $(x + 4)^2$

$(x + 4)^2 = x^2 + 8x + 16$, so subtract 16

$x^2 + 8x + 10 = (x + 4)^2 + 10 - 16 = (x + 4)^2 - 6$

$\boxed{(x + 4)^2 - 6}$

The vertex is at (-4, -6).

Example 2 – negative coefficient of x

Write $x^2 - 6x + 2$ in completed square form.

Half of -6 is -3, so bracket is $(x - 3)^2$

$(x - 3)^2 = x^2 - 6x + 9$, so subtract 9

$x^2 - 6x + 2 = (x - 3)^2 + 2 - 9 = (x - 3)^2 - 7$

$\boxed{(x - 3)^2 - 7}$

The minimum value of this expression is -7.

Example 3 – no constant term

Write $x^2 + 10x$ in completed square form.

Half of 10 is 5, so bracket is $(x + 5)^2$

$(x + 5)^2 = x^2 + 10x + 25$

$x^2 + 10x = (x + 5)^2 - 25$

$\boxed{(x + 5)^2 - 25}$

When c = 0, q equals the negative of p^2.

Example 4 – solving by completing the square

Solve $x^2 + 4x - 5 = 0$ by completing the square.

Complete: $(x + 2)^2 - 4 - 5 = 0$ $(x + 2)^2 = 9$

$x + 2 = \pm 3$

$x = 1$ or $x = -5$

$\boxed{x = 1 \text{ or } x = -5}$

Take the square root of both sides, remembering ±.

YOUR TURN – PRACTICE (deliberate practice)

1. Positive x coefficient
Write in completed square form $(x + p)^2 + q$.

a) $x^2 + 4x + 1$

b) $x^2 + 6x + 7$

c) $x^2 + 10x + 20$

d) $x^2 + 2x - 3$

2. Negative x coefficient
Write in completed square form.

a) $x^2 - 4x + 3$

b) $x^2 - 8x + 10$

c) $x^2 - 2x - 5$

d) $x^2 - 12x + 30$

3. No constant term
Complete the square.

a) $x^2 + 6x$

b) $x^2 - 10x$

c) $x^2 + 14x$

d) $x^2 - 20x$

4. Solving equations
Solve by completing the square.

a) $x^2 + 6x + 5 = 0$

b) $x^2 - 4x - 12 = 0$

c) $x^2 + 2x - 8 = 0$

d) $x^2 - 10x + 9 = 0$

5. Finding the vertex
Complete the square and state the minimum point (vertex).

a) $x^2 + 4x + 7$ b) $x^2 - 6x + 11$ c) $x^2 + 8x + 20$ d) $x^2 - 2x - 3$ e) $x^2 + 12x + 32$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) To complete the square, halve the coefficient of x ☐

Correct answer: __________

Reason: __________________

b) $x^2 + 6x + 5 = (x + 6)^2 + 5$ ☐

Correct answer: __________

Reason: __________________

c) $(x + 3)^2 = x^2 + 6x + 9$ ☐

Correct answer: __________

Reason: __________________

d) $x^2 + 6x + 5 = (x + 3)^2 + 5$ ☐

Correct answer: __________

Reason: __________________

e) The vertex of $(x + p)^2 + q$ is at $(-p, q)$ ☐

Correct answer: __________

Reason: __________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Write $\mathbf{2x^2 + 12x + 7}$ in the form $\mathbf{a(x + p)^2 + q}$. (Hint: factor out 2 first.) State the coordinates of the vertex.

Expression: ______________________

Simplified: ______________________

b) The equation $\mathbf{x^2 + kx + 9 = 0}$ has exactly one solution. Find the possible values of k by completing the square and setting the discriminant to zero.

Expression: ______________________

Simplified: ______________________

TOP TIPS
Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK
Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!